

Table 1: Statistics of test sets in this paper

Category	Task	Original Prompt
Semantics	Sentiment Analysis (100)	Determine whether a movie review is positive or negative.
	Sentence Similarity (100)	Rate the semantic similarity of two input sentences on a scale of 0 - definitely not to 5 - perfectly.
	Word in Context (100)	Determine whether an input word has the same meaning in the two input sentences.
	Cause Selection (25)	Find which of the two given cause and effect sentences is the cause.
Knowledge	Larger Animal (100)	Write the larger of the two given animals.
Numerical	Sum (100)	Sum the two given numbers.
Spelling	Starting With (100)	Extract the words starting with a given letter from the input sentence.
	First Letter (100)	Extract the first letter of the input word.

Instruction Induction [Honovich *et al.*, 2022], which is designed by humans, the other is Zero-shot-CoT [Kojima *et al.*, 2022]. To our knowledge, it is the simplest and efficient approach in zero-shot prompt engineering.

Datasets and Tasks We access the efficiency of our approach on eight tasks of Instruction Induction [Honovich *et al.*, 2022]: Sentiment Analysis (SA), Sentence Similarity (SS), Cause Selection (CS), Sum, Word in Context (WC), Starting With (SW), Larger Animal (LA) and First Letter (FL). Details on those tasks can be found in Table 1. For each task, we test on 100 samples, except for Cause Selection, including 50 examples in total. And we choose in-context demonstrations for few-shot learning from the remaining part of the data.

4.2 Our Emotional Prompts

Based on three widely known psychology theories, we design 11 emotional stimuli to enhance the performance of LLMs; details are shown in Table 2. As shown in Fig. 2, the emotion stimulus 02~05 stem from Social Identity theory [Hogg, 2016; Turner and Oakes, 1986], 07~11 are derived from Cognitive Emotion Regulation theory [Barańczuk, 2019], and

Table 2: Our Emotion Stimulus

Stimulus ID	Stimulus Content
EP_01	, write your answer and give me a confidence score between 0-1 for your answer.
EP_02	This is very important to my career.
EP_03	You’d better be sure.
EP_04	Are you sure?
EP_05	Are you sure that’s your final answer? It might be worth taking another look.
EP_06	Provide your answer and a confidence score between 0-1 for your prediction. Additionally, briefly explain the main reasons supporting your classification decision to help me understand your thought process. This task is vital to my career, and I greatly value your thorough analysis.
EP_07	Are you sure that’s your final answer? Believe in your abilities and strive for excellence. Your hard work will yield remarkable results.
EP_08	Embrace challenges as opportunities for growth. Each obstacle you overcome brings you closer to success.
EP_09	Stay focused and dedicated to your goals. Your consistent efforts will lead to outstanding achievements.
EP_10	Take pride in your work and give it your best. Your commitment to excellence sets you apart.
EP_11	Remember that progress is made one step at a time. Stay determined and keep moving forward.

And 01~04 conform to Social Cognition theory [Fiske and Taylor, 1991; Luszczynska and Schwarzer, 2015]. Note that “EP_06” is a compound stimulus, which combines “EP_01”, “EP_02” and “EP_03”.

4.3 Main Results

The main results are shown in Table 3. We see that our EmotionPrompt achieves comparable or even better performance than the original one. We now list the specific findings on each LLM:

1. **ChatGPT:** EmotionPrompt achieves better or comparable performance on all tasks in zero-shot and few-shot settings. The accuracy increased over 10% on 4 and 5 tasks in zero-shot and few-shot scenarios, respectively.
2. **Flan-T5-Large:** EmotionPrompt outperforms original prompt and Zero-Shot-CoT on 6 tasks and 5 tasks in zero-shot scenario. For few-shot setting, it outperforms original prompt and Zero-Shot-CoT on six tasks, and all prompts fails on the last two tasks.

Table 3: Results on Different Models and Tasks. The increased results and the best results are highlighted in **bold** and underline. The value 0.00 indicates that we do not receive meaningful or useful response that is left for future investigation.

Model	Prompt	Zero-shot								Few-shot							
		SA	SS	LA	Sum	SW	WC	CS	FL	SA	SS	LA	Sum	SW	WC	CS	FL
ChatGPT	Origin	0.82	0.41	0.77	0.93	0.32	0.51	0.96	1.00	0.78	0.4	0.9	1.00	0.41	0.52	0.72	1.00
	EP_01	0.91	0.52	0.91	1.00	0.36	0.54	1.00	1.00	0.9	0.39	0.9	1.00	0.2	0.57	0.64	1.00
	EP_02	0.85	0.53	0.91	1.00	0.46	0.57	0.96	1.00	0.82	0.47	0.92	1.00	0.54	0.52	0.8	1.00
	EP_03	0.86	0.52	0.9	1.00	0.48	0.57	1.00	1.00	0.82	0.42	0.93	1.00	0.45	0.59	0.76	1.00
	EP_04	0.83	0.49	0.88	1.00	0.45	0.51	0.96	1.00	0.82	0.44	0.9	1.00	0.42	0.56	0.72	1.00
	EP_05	0.84	0.5	0.88	1.00	0.49	0.55	0.96	1.00	0.84	0.42	0.95	1.00	0.46	0.55	0.56	1.00
	EP_06	0.89	0.56	0.91	1.00	0.32	0.62	1.00	1.00	0.95	0.39	0.87	1.00	0.35	0.55	0.64	1.00
	EP_07	0.88	0.49	0.83	1.00	0.38	0.47	0.88	1.00	0.8	0.42	0.94	1.00	0.47	0.49	0.88	1.00
	EP_08	0.86	0.51	0.9	1.00	0.43	0.63	0.64	1.00	0.84	0.44	0.93	1.00	0.49	0.62	0.84	1.00
	EP_09	0.87	0.52	0.88	1.00	0.53	0.54	0.8	1.00	0.8	0.48	0.9	1.00	0.51	0.56	0.8	1.00
	EP_10	0.87	0.56	0.91	1.00	0.49	0.58	0.72	1.00	0.77	0.41	0.91	1.00	0.47	0.59	0.88	1.00
	EP_11	0.88	0.54	0.9	1.00	0.49	0.54	0.96	1.00	0.84	0.56	0.93	1.00	0.47	0.57	0.8	1.00
	avg	0.87	0.52	0.89	1.00	0.44	0.56	0.90	1.00	0.84	0.44	0.92	1.00	0.44	0.56	0.76	1.00
	CoT	0.79	0.47	0.91	1.00	0.31	0.52	0.96	1.00	0.77	0.52	0.92	1.00	0.42	0.52	0.76	1.00
T5	Origin	0.93	0.55	0.71	0.57	0.56	0.03	0.03	0.00	0.92	0.54	0.61	0.03	0.00	0.47	0.52	0.00
	EP_01	0.91	0.64	0.52	0.41	0.00	0.06	0.02	0.00	0.93	0.54	0.63	0.04	0.00	0.43	0.00	0.00
	EP_02	0.93	0.58	0.71	0.58	0.48	0.04	0.02	0.00	0.92	0.56	0.62	0.04	0.00	0.51	0.68	0.00
	EP_03	0.93	0.57	0.69	0.61	0.72	0.05	0.03	0.00	0.93	0.58	0.65	0.03	0.00	0.53	0.68	0.00
	EP_04	0.93	0.58	0.58	0.57	0.64	0.07	0.02	0.00	0.93	0.56	0.63	0.03	0.00	0.51	0.68	0.00
	EP_05	0.92	0.57	0.74	0.58	0.6	0.08	0.03	0.00	0.93	0.58	0.69	0.02	0.00	0.49	0.72	0.00
	EP_06	0.97	0.61	0.55	0.44	0.00	0.11	0.03	0.00	0.93	0.54	0.71	0.03	0.00	0.49	0.04	0.00
	EP_07	0.92	0.58	0.68	0.6	0.48	0.08	0.02	0.00	0.93	0.55	0.66	0.03	0.00	0.48	0.56	0.00
	EP_08	0.93	0.58	0.64	0.56	0.64	0.08	0.02	0.00	0.93	0.6	0.61	0.04	0.00	0.5	0.64	0.00
	EP_09	0.93	0.58	0.64	0.57	0.44	0.08	0.02	0.00	0.93	0.58	0.63	0.04	0.00	0.51	0.52	0.00
	EP_10	0.93	0.58	0.62	0.59	0.48	0.09	0.02	0.00	0.92	0.58	0.6	0.04	0.00	0.48	0.6	0.00
	EP_11	0.93	0.59	0.64	0.55	0.56	0.07	0.02	0.00	0.93	0.57	0.64	0.04	0.00	0.49	0.64	0.00
	avg	0.93	0.59	0.64	0.55	0.46	0.07	0.02	0.00	0.93	0.57	0.64	0.03	0.00	0.49	0.52	0.00
	CoT	0.94	0.58	0.87	0.51	0.52	0.06	0.03	0.00	0.92	0.54	0.6	0.03	0.00	0.45	0.48	0.00
Bloom	Origin	0.68	0.23	0.74	0.41	0.06	0.52	0.48	0.86	0.77	0.24	0.43	0.07	0.18	0.52	0.6	0.98
	EP_01	0.76	0.26	0.6	0.18	0.03	0.42	0.52	0.96	0.91	0.26	0.52	0.01	0.22	0.6	0.76	0.99
	EP_02	0.79	0.2	0.75	0.45	0.08	0.5	0.6	0.84	0.87	0.23	0.54	0.12	0.19	0.64	0.72	0.99
	EP_03	0.72	0.23	0.81	0.37	0.05	0.51	0.52	0.85	0.85	0.23	0.49	0.06	0.2	0.56	0.6	1.00
	EP_04	0.66	0.28	0.67	0.35	0.03	0.5	0.36	0.88	0.83	0.3	0.52	0.03	0.22	0.52	0.8	1.00
	EP_05	0.81	0.21	0.71	0.27	0.02	0.57	0.48	0.85	0.85	0.27	0.51	0.04	0.16	0.53	0.76	0.99
	EP_06	0.56	0.2	0.74	0.14	0.09	0.24	0.4	0.94	0.89	0.21	0.46	0.01	0.23	0.59	0.64	1.00
	EP_07	0.64	0.3	0.7	0.34	0.02	0.48	0.72	0.71	0.86	0.3	0.54	0.09	0.22	0.54	0.68	0.98
	EP_08	0.4	0.23	0.72	0.35	0.01	0.47	0.48	0.52	0.85	0.27	0.51	0.08	0.18	0.6	0.76	0.98
	EP_09	0.38	0.19	0.79	0.39	0.04	0.41	0.8	0.76	0.87	0.24	0.48	0.11	0.22	0.56	0.76	0.98
	EP_10	0.44	0.22	0.76	0.42	0.03	0.46	0.44	0.68	0.87	0.28	0.47	0.12	0.19	0.53	0.8	1.00
	EP_11	0.62	0.24	0.79	0.4	0.05	0.44	0.68	0.86	0.9	0.21	0.53	0.07	0.17	0.56	0.68	0.99
	avg	0.62	0.23	0.73	0.33	0.04	0.45	0.55	0.80	0.87	0.25	0.51	0.07	0.2	0.57	0.72	0.99
	CoT	0.57	0.22	0.79	0.49	0.03	0.44	0.6	0.86	0.88	0.2	0.46	0.06	0.16	0.54	0.68	0.99
Vicuna	Origin	0.4	0.24	0.82	0.41	0.02	0.46	0.56	0.00	0.77	0.28	0.68	0.32	0.05	0.47	0.32	0.00
	EP_01	0.60	0.25	0.84	0.90	0.00	0.44	0.60	0.00	0.53	0.33	0.46	0.29	0.05	0.53	0.44	0.00
	EP_02	0.28	0.25	0.82	0.61	0.06	0.57	0.06	0.00	0.8	0.32	0.60	0.35	0.09	0.51	0.52	0.00
	EP_03	0.49	0.32	0.85	0.76	0.02	0.47	0.56	0.00	0.76	0.36	0.58	0.32	0.03	0.45	0.44	0.00
	EP_04	0.3	0.28	0.59	0.59	0.02	0.51	0.64	0.00	0.69	0.24	0.54	0.05	0.04	0.51	0.52	0.00
	EP_05	0.50	0.25	0.64	0.40	0.00	0.50	0.48	0.00	0.70	0.29	0.26	0.05	0.01	0.31	0.48	0.00
	EP_06	0.71	0.31	0.83	0.81	0.01	0.55	0.76	0.00	0.39	0.27	0.29	0.12	0.03	0.45	0.32	0.00
	EP_07	0.41	0.20	0.87	0.84	0.03	0.50	0.64	0.00	0.66	0.27	0.63	0.13	0.02	0.34	0.48	0.00
	EP_08	0.34	0.25	0.91	0.89	0.02	0.55	0.76	0.00	0.81	0.3	0.71	0.48	0.05	0.53	0.48	0.00
	EP_09	0.41	0.28	0.87	0.87	0.04	0.55	0.72	0.00	0.76	0.37	0.71	0.41	0.07	0.46	0.48	0.00
	EP_10	0.43	0.25	0.87	0.82	0.05	0.53	0.76	0.00	0.78	0.35	0.68	0.43	0.07	0.50	0.48	0.00
	EP_11	0.3	0.27	0.89	0.17	0.05	0.47	0.68	0.00	0.82	0.24	0.65	0.43	0.08	0.48	0.44	0.00
	avg	0.43	0.26	0.82	0.70	0.03	0.51	0.65	0.00	0.70	0.30	0.56	0.28	0.05	0.46	0.46	0.00
	CoT	0.34	0.28	0.74	0.02	0.01	0.42	0.4	0.00	0.8	0.24	0.42	0.44	0.05	0.50	0.37	0.00

3. **Vicuan-13b**: EmotionPrompt performs better on all tasks in zero-shot and few-shot settings. Specifically

for zero-shot setting, EmotionPrompt increases the accuracy on Sum from 0.41 to 0.90, on Sentiment Analysis from 0.40 to 0.71, and on Cause Selection from 0.56 to 0.76.

4. **Bloom:** EmotionPrompt outperforms original prompts on all tasks. The average values of EmotionPrompt are higher than original prompts on 7/8 tasks in few-shot.

4.4 Truthfulness and Informativeness

We employ EmotionPrompt on TruthfulQA [Lin *et al.*, 2021] to explore the effect of EmotionPrompt on truthfulness and informativeness. The benchmark has 817 questions that span 38 categories, including health, law, finance and politics. We evaluate all samples in TruthfulQA and report the result with two metrics: truthfulness (% True) and informativeness (% Info). Those results can be accessed by their fine-tuned GPT-judge and GPT-info, which have been proven to align with human prediction over 90% of the time.

Table 5 shows the results on ChatGPT, Vicuna-13b and T5. EmotionPrompt increases the truthfulness on ChatGPT from 0.75 to 0.87, Vicuna-13b from 0.77 to 1.0 and T5 from 0.54 to 0.77. EmotionPrompt also improves the informativeness on ChatGPT from 0.53 to 0.94 and T5 from 0.42 to 0.48.

4.5 Effect of More Emotional Stimulus

As one or more stimuli may regulate human action, and more stimuli sometimes are more efficient, we explore the effect of more emotional stimuli on LLMs. We randomly combine some emotional stimuli and experiment on ChatGPT; results are shown in Table 6. Our findings are listed below:

1. **More emotional stimuli generally lead to better performance in most cases.** The second group and the third group explore the effect of adding “EP_01”, showing that the third group performs better than the second group in most cases.
2. **Combined stimulus can bring little or no benefit when sole stimulus already achieves a good performance.** The combination “EP_01” + “EP_04” gets a high score in most tasks and does not improve significantly or even decrease when we add more stimulus to it, such as stimulus 06~09.

5 Analysis

5.1 Why EmotionPrompt Works?

EmotionPrompt shows great improvements on eight tasks not only in accuracy but also in truthfulness and informativeness. Why does EmotionPrompt work? We explain this by visualizing the emotional stimulus’s input attention contributions to the final outputs as proposed in [Zhu *et al.*, 2023]. Because T5-large is open-sourced and relatively small, we chose it as our experimental LLM and accessed every word’s contribution based on the gradient norm. The experiment is conducted on a Sentiment Analysis task. We compute prompts’ contributions on every test sample and use the average value to represent their importance.

Table 4 shows every word’s contribution to the final result, and the color depth represents their importance. Our key findings are as follows:

1. **Emotional stimulus can enhance original prompts’ representation.** Original prompt “Determine whether a movie review is positive and negative.” have deeper color in EmotionPrompt, especially in “EP_01”, “EP_03”, “EP_06”, “EP_07”, “EP_08”, “EP_09” and “EP_10”. This means emotional stimulus can enhance the attention of original prompts.
2. **Positive words make more contributions.** In our designed emotional stimulus, some positive words play a more important role, such as “confidence”, “sure”, “success” and “achievement”. Based on this finding, we summarize positive words’ contributions and their total contributions to the final result on eight tasks. As shown in Fig. 3, the contributions of positive words pass 50% on four tasks, even approach 70% on two tasks.

5.2 Human Study

To explore the effect of EmotionPrompt in more aspects, we conduct a human study to access other metrics of LLMs’ outputs, such as clarity, relevance (with the question), depth, structure and organization, supporting evidence and engagement (with humans), which are designed based on Elaboration Likelihood model [Petty and Briñol, 2011], Cognitive Load Theory [Sweller, 2011] and Schema theory [McVee *et al.*, 2005]. We prepared 40 problems from TruthfulQA [Lin *et al.*, 2021] and add emotional stimulus “EP_04”, “EP_06”, “EP_11” to original prompts, then we got the outputs of the original prompt and three EmotionPrompt via ChatGPT. Four volunteers rate four outputs on a scale of 1 - bad to 5 - good on six metrics for 40 questions. We report the average values in Table 7. Results show that EmotionPrompt access better performance on clarity, depth, structure, and organization, supporting evidence and engagement.

6 Conclusion and Limitation

We have proposed EmotionPrompt, which enhances the performance of LLMs via emotional stimulus. Our method is not only simple but also powerful; method achieves awesome performance on eight tasks with four LLMs. Our study to explore the emotional intelligence of LLMs might encourage the community to further examine the emotional power of LLMs and other emotional stimuli.

This work has several limitations. First, we only experiment with four LLMs and conduct experiments in several tasks with few test examples, which are limited. Thus, our conclusions about emotion stimulus can only work on our experiments and any LLMs and datasets out of the scope of this paper might not work with emotion stimulus. Second, the emotional stimulus proposed in this paper may not be general to other tasks, and researchers may propose other useful replacements for your own tasks. Finally, since results may change due to the version change of ChatGPT, we cannot ensure its reproducibility.

Ethical Statement

In this study, we aim to assess the ability of LLMs to understand emotion. While various ethical issues are linked